

Removing Digits II

Time limit: 1.00 s

Memory limit: 512 MB

You are given an integer n . On each step, you may subtract from it any one-digit number that appears in it. How many steps are required to make the number equal to 0?

Input

The only input line has an integer n .

Output

Print one integer: the minimum number of steps.

Constraints

- $1 \leq n \leq 10^{18}$

Example

Input	Output
27	5

Explanation: An optimal solution is $27 \rightarrow 20 \rightarrow 18 \rightarrow 10 \rightarrow 9 \rightarrow 0$.